

ECO401-INTRODUCTION TO ECONOMICS

Final term Short notes by

MR Shāhīzāz

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Final term exams of the current semester Fall 2023 will comprise of the contents from Lecture 19 to 45. This paper will cover both objective (MCQs) and subjective type questions (3 and 5 Marks). So, kindly prepare your exam accordingly.

Lesson 19 Market Structures (Continued)

Perfect Competition (Long Run):

- All factors variable, firms can enter/exit.
- Supernormal profits attract, increasing supply.
- Losses lead to exits, reducing supply.
- Perfectly competitive firms earn normal profits in the long run.

Allocative & Productive Efficiency:

- Allocative: $MR=MC$ for individual firms, $P=MC$ for society. Achieved in perfect competition.
- Productive: $P=MC=AC$. Achieved by perfectly competitive firms in the long run.

Monopoly:

- Single producer in the market.
- Monopoly power inversely related to price elasticity of demand.

Profit Maximization under Monopoly:

- $MC=MR$ for profit maximization.
- Demand curve downward-sloping, unlike perfect competition.

Retaining Monopoly:

- Natural reasons or active policies.
- Economies of scale, product differentiation, brand loyalty.
- Active pricing strategies (limit pricing).
- Threat of takeover, control of key factors, patents or copyrights.

Lesson 20 Market Structures (Continued)

Limit Pricing:

- Established firms, like monopolies, exploit the cost advantage over new entrants.
- Monopolist charges price PM to maximize profits at output QM .
- To deter new entrants, monopolist can set a price lower than PM .

Monopolies and Public Interest:

- Disadvantages:
 - Produce lower quantities at higher prices than perfectly competitive firms.
 - Earn supernormal profits.
 - Most surplus benefits the monopolists.
 - Lack sufficient incentive to increase production efficiency.

Equilibrium under Perfect Competition and Monopoly:

- Monopolists do not produce where $P=MC$ or $P=AC$ minimum.
- Higher prices and lower quantities compared to perfect competition.

Advantages of Monopolies:

- Natural Monopolies are efficient for society.
- Supernormal profits can fund R&D, innovation, and expansion into foreign markets.

Government Regulation:

- Government can regulate monopolies to set prices at the intersection of AR and MC for allocative efficiency.
- Ensuring productive efficiency is challenging, and subsidies or taxes may be required.
- Governments may regulate at the point where AR intersects AC to approach efficiency without subsidies or taxes.

Lesson 21 Market Structures (Continued)

Price Discrimination (PD):

Types of PD:

- 1st Degree PD:** Charging different prices based on what each customer can pay. Example: new and used car sales.
- 2nd Degree PD:** Charging different prices for different quantities. Examples: last-minute hotel and airline deals, retail discounts for buying multiple items.

3. 3rd Degree PD: Charging different prices to different customers in different markets.
Examples: international pricing by fast-food chains, discounts for seniors or students.

Consequences of PD:

- Allows firms to make profits or increase supernormal profits.
- Enables production of goods that might not be produced otherwise.

Pre-requisites/Conditions of Price Discrimination:

- i. Independent markets.
- ii. Firms with pricing flexibility (not in perfect competition).
- iii. Different price elasticities of demand for various customers.

Benefits of Price Discrimination:

- Can be beneficial or harmful for public interest.
- Depends on equity concerns, production of unique goods, and use of supernormal profits.

Monopolistic Competition:

- Characteristics similar to perfect competition, with differentiated products.
- Firms are price-takers, but not to the extreme extent of perfect competition.
- Short-run supernormal profits, long-run normal profits.
- Advertising and product promotion play a role due to product differentiation.

Short Run and Long Run under Monopolistic Competition:

- Short-run supernormal profits possible.
- Long-run equilibrium where AR curve is tangent to AC curve.
- Public interest depends on the position of AC at the point of tangency.
- Room for advertising and product promotion due to product differentiation.

Lesson 22: Market Structures (Continued) –

Oligopoly, Collusion, and Game Theory

Oligopoly:

- Similar to monopoly due to a small number of firms (2-20) and barriers to entry.
- Resembles perfect competition in competitive behavior.
- Shares similarities with monopolistic competition, allowing for differentiated products.

Characteristics of Oligopoly:

- Few participants lead to interdependence.
- Strategic planning involves considering others' likely responses.
- No single equilibrium due to firms being interdependent.

Collusion:

- Occurs when firms cooperate in setting prices or quantities.
- Aims to maximize industry profits while individual firms seek personal gains.
- Common in oligopolies, with cartels being a special form of explicit collusion.
- Tacit collusion occurs when cooperation is implicit, not explicit.

Two Oligopoly Scenarios:

1. Collusive Oligopoly (Cartel):

- Formed through agreements on market shares, prices, advertising, or quotas.
- Example: OPEC.
- Cartels are most likely to survive under certain conditions.

2. Non-Collusive Oligopoly:

- Firms do not cooperate, and collusion breaks down.
- High incentive to cheat on agreements.
- Oligopolies often oscillate between collusive and non-collusive states.

Collusion and Game Theory:

- Prisoner's Dilemma: A scenario where rational, self-interested choices lead to a suboptimal outcome.
- Applies to oligopolistic firms when independent strategic decisions result in a worse position than cooperation.
- Payoff matrix illustrates the outcomes based on different pricing strategies.

Strategic Decision-Making:

- **Maximin Strategy:** Maximizes the minimum profit, a cautious approach.
- **Maximax Strategy:** Maximizes the maximum payoff, an optimistic approach.

- **Dominant Strategy Game:** Both firms' dominant strategy leads to a suboptimal outcome, emphasizing the benefits of collusion.

Key Takeaways:

- Collusion in oligopolies can lead to both positive and negative outcomes.
- Understanding game theory helps firms strategize in competitive markets.
- Dominant strategies may not always lead to the best collective outcome.
- Balancing self-interest and cooperation is crucial in oligopolistic markets.

Lesson 23: Market Structures (Continued) & Welfare Economics

Price Stability in Non-Collusive Oligopolies: Kinked Demand Curve:

- Explains price "stickiness" in oligopolistic markets.
- Assumptions: If one firm raises prices, others won't follow (elastic demand), and if one lowers prices, others will follow (inelastic demand).
- Kinked demand curve has two segments: more elastic for price increases and less elastic for price decreases.

Non-Price Competition:

- Competition based on factors other than price, e.g., advertising expenditures.

Oligopoly & Public Interests:

- Oligopolies can earn supernormal profits, similar to monopoly.

- Firms can use profits for cost minimization.

Welfare Economics:

- Branch dealing with normative issues, determining what should be.
- Analyzes allocative efficiency and income distribution in an economy.
- Focuses on social welfare and economic activities.

Marginal Private Cost of Advertising:

- Cost of additional advertising borne by a firm.
- Does not include nuisance costs to viewers/readers.
- A welfare economics concern.

Marginal Social Cost:

- Social cost includes costs borne by society due to production.
- Marginal social cost = marginal private costs + other costs borne by society.

Externality:

- Occurs when the production or consumption of a good affects those not involved in buying/selling.
- Positive (beneficial) or negative (harmful) externality.
- Production externality vs. consumption externality.

Optimal Level of Production:

- Socially optimal level internalizes the externality.
- Equilibrium at the intersection of marginal social benefit and cost curves.
- Not necessarily at the intersection of marginal private benefit and cost curves.

Merit Good:

- A good the government encourages people to consume more.
- Determined by societal norms, not just consumer preferences.
- Examples include education, health care, and subsidized housing.

Public Good:

- Benefits spread among the entire community.
- Characteristics: non-rivalry and non-excludability.
- Examples: national defense, clean air, public fireworks, etc.

Lesson 24: Welfare Economics (Continued) & Economics of Information Products

Market for Factors of Production:

- Circular flow shows the flow of goods and factors between households and firms.
- Firms demand factors, and households supply them.
- The demand for factors is derived from the goods market.

Leisure and Marginal Disutility of Work (MDUW):

- Leisure is non-working time used for relaxation.
- MDUW is the negative impact on a laborer for an additional unit of work.
- Labor supply curve is positively sloped but may bend backward due to MDUW.

Opportunity Cost and Supply-Demand Curve for Labor:

- Opportunity cost of working is leisure.
- Labor supply curve may bend backward above a certain wage rate.
- Wage rate is the marginal cost of labor, derived from the intersection of wage rate lines and marginal revenue product of labor.

Value of Marginal Product of Labor (VMPL):

- Represents the additional value generated by an additional unit of labor.
- $VMPL = MPPL \times Price (P_i)$.
- In perfect competition, $VMPL = MRPL$, but in other cases, $VMPL > MRPL$.
- Labor can be rented, unlike land and capital which can be purchased.

Net Present Value (NPV) and Discounting:

- NPV is the discounted value of net returns and disposal value minus the initial purchase cost.
- Discounting converts future incomes and expenses into present value.
- The discount rate determines the rate at which future incomes are discounted.

Economics of Information Products:

- Studies how economic principles apply to the production, distribution, and consumption of information products.
- Internet has reduced the marginal cost of distributing information to zero.
- Information is non-rivalrous and has almost zero marginal cost, making classic marginal cost pricing infeasible.

Experience Goods:

- Goods where characteristics such as quality or price are difficult to observe in advance.
- Consumers need to experience the product before making a decision.
- Examples include services or products where quality is subjective and requires firsthand experience.

Lesson 25: Introduction to Macroeconomics

Macroeconomics:

- Emerged in the 1940s, influenced by economist John Maynard Keynes.
- Focuses on the national economy as a whole, studying its performance, structure, and behavior.
- Variables shift from micro-level product-specific factors to economy-wide factors like aggregate demand and aggregate supply.

Aggregate Demand (AD):

- Total planned or desired spending in the economy over a period.
- Comprises consumption, investment, government spending, and net exports.

- Inversely related to the aggregate price level.

Aggregate Supply (AS):

- The total value of goods and services all firms can and would willingly produce in a given period.
- A function of inputs, technology, and price level.
- Slopes upward in P-Output space, influenced by whether the economy operates below full employment (flat) or at full employment (steep).

Classical Economics:

- Developed by Adam Smith, David Ricardo, Thomas Malthus, and John Stuart Mill.
- Microeconomists who considered macroeconomics as an aggregation of micro-markets.
- Optimal role of government: Laissez-faire, allowing the market forces to determine prices and resource allocation.

The Concept of Invisible Hand:

- Introduced by Adam Smith in 1776.
- Describes the paradox of a laissez-faire market economy.
- Suggests that, with participants pursuing self-interest, the market system benefits all as if guided by an invisible hand.
- Believes in the efficiency of perfectly competitive markets.

Full Employment:

- Classical economists assumed the economy, left to itself, would tend to full employment equilibrium.

- Full employment defined as the state where productive resources are fully employed.
- A state where no or minimal involuntary unemployment exists.

Classical Views About Great Depression:

- Low investment, high unemployment, and low output were viewed as temporary problems.
- Recommended increasing savings to lower interest rates and boost investment.
- Advocated removing obstructions in the labor market to allow wages to fall to market-clearing levels.

Failure of the Classical Model:

- The classical model failed after 1930, especially during the Great Depression.
- Did not provide a satisfactory solution to low investment, high unemployment, and falling output.
- Focused on removing impediments to the free market without addressing broader issues.

Keynes and the Origins of Modern Macroeconomics:

- Keynesian economics, based on the ideas of John Maynard Keynes, emerged as a response to the Great Depression.
- Promotes a mixed economy where both the state and the private sector play important roles.
- Advocates for government policies to promote demand at a macro level and fight high unemployment and deflation.

Keynesians' Views About Great Depression:

- Keynes argued that low demand and static pessimistic expectations were the overarching problems.

- Recommended increasing household consumption to boost demand.
- Believed unemployment was rising due to market equilibrium moving away from full employment, not wage-related issues.

Lesson 26: Introduction to Macroeconomics (Continued)

Differences Between Classical and Keynesian Economics:

1. Labor Market:

- **Classical View:**
 - Excess supply of labor causes wages to fall, increasing labor demand and restoring market equilibrium.
 - Assumes wages are flexible.
- **Keynesian View:**
 - Firms view falling wages negatively as it reduces consumers' purchasing power, leading to lower investment.
 - Wages are sticky downwards, don't fall easily, causing prolonged unemployment.

2. Market for Loanable Funds:

- **Classical View:**
 - Higher interest rates hindered investment during the Great Depression.
 - Encouraged increased savings to lower interest rates and stimulate investment.
- **Keynesian View:**
 - Increased savings reduce consumption and, subsequently, demand for new investment.

- A fall in investment demand leads to a new market clearing interest rate.

3. Aggregate Demand and Supply:

- Both Keynesian and classical views have a downward-sloping aggregate demand (AD) curve.

- The AD curve slopes downward due to three effects:

1. **Interest Rate Effect:** Higher prices lead to a rise in nominal interest rates, discouraging firm investment.
2. **Wealth Effect:** Higher prices reduce the purchasing power of consumers' income and wealth, decreasing consumption demand.
3. **International Purchasing Power Effect:** Higher prices make exports more expensive, reducing demand in international markets.

Factors Shifting Aggregate Demand:

- AD shifts right when any component increases autonomously:
- Consumers become more willing to spend at every price level.
- Autonomous increases in investment due to improved business prospects.
- Government increases spending or reduces taxes.
- Net exports rise at all prices due to factors like improved quality of domestic goods relative to foreign goods.

Summary of Keynes' Views:

- Keynes believed an economy could settle below full employment.
- Advocated demand-side policies:
- Government spending and encouraging consumption to lift the economy out of depression.

- A virtuous circle of increased demand, production, hiring, and further increased demand would change agents' pessimistic views and lead to recovery.

Lesson 27 Introduction To Macroeconomics (Continued)

1. Classical View of Aggregate Supply (AS):

- AS curve is vertical, suggesting full employment.
- Lack or excess of demand doesn't affect output.
- Policy recommendation: Focus on shifting AS curve right (supply side measures).
- Economy automatically finds equilibrium in the long run.

2. Keynesian View of Aggregate Supply (AS):

- AS curve is horizontal at less than full employment, upward sloping afterwards.
- Injection of demand can increase output, employment, and income.
- Short run: Prices fixed, output increases with demand.
- Long run: AS curve becomes vertical.

3. Keynesian Demand Management Policies:

- Used during economic fluctuations.
- Government intervenes to increase demand in recessions and decrease it in booms.
- Major problem: Couldn't handle stagflation (inflation and unemployment rising simultaneously).

4. Different Schools of Thought:

- **Monetarist School:** Inflation tied to monetary growth.
- **Real Business Cycles (RBC) School:** Output fluctuations due to technology shocks.
- **Rational Expectations School:** Agents use all available information in decision-making, predictable policies ineffective.
- **Neo-Classical Economics:** Reinforces free market principles, stresses micro-foundations.
- **Neo-Keynesian School:** Highlights market failures, suggests meaningful government intervention due to information asymmetries and coordination failures.

Lesson 28 Macroeconomic Data and National Income Accounting

Understanding Macroeconomic Data:

1. **Selective Use of Data:** Data can be selectively used, ignoring certain information.
2. **Graphical Presentation:** Scales in graphs can manipulate the perception of data.
3. **Absolute vs. Proportionate Values:** Absolute values might not reflect changes accurately.
4. **Distribution Considerations:** Changes in wealth distribution may not be reflected in overall economic growth.
5. **Nominal vs. Real Data:** Nominal data is unadjusted for inflation, real data is adjusted.
6. **Exclusion of Time Periods:** Selective time frames may skew interpretations.
7. **Aggregate vs. Per Capita Data:** National income might increase while per capita income decreases due to population growth.
8. **Manipulation of Growth Rates:** Growth rates can be influenced by population changes.

National Income Accounting:

1. **Gross Domestic Product (GDP):** Total market value of all goods and services produced within a country's borders in a given period.
2. **Flow vs. Stock:** Flow refers to a period of time, stock refers to a point in time.
3. **Methods of Measuring GDP:** Product (value added) method, expenditure method, factor income method.
4. **Value Added:** Difference between the value of goods produced and the cost of materials used.
5. **Final vs. Intermediate Goods:** Final goods are for end use, intermediate goods are inputs for further production.
6. **Calculation of GDP:** GDP is the sum of value added at each stage of production, avoiding double counting.
7. **Market Prices vs. Factor Cost:** GDP can be calculated at market prices (including sales tax) or at factor cost (excluding sales tax).

Lesson 29 Macroeconomic Data and National Income Accounting (Continued)

Macroeconomic Data & National Income Accounting:

- **GDP at Factor and Market Prices:**
 - Factor price: Price at which a firm sells its final output to consumers.
 - Market price: Includes factor price plus indirect taxes imposed by the government.
 - GDP at market price is higher than GDP at factor price.
- **Net Domestic Product (NDP):**
 - Obtained by subtracting depreciation from GDP.

- Depreciation: Reduction in the value of capital goods due to wear and tear during production.

- **Gross National Product (GNP):**
 - Value of all final goods and services produced during a year by factors owned by citizens.
 - $GNP = GDP + \text{net factor income from abroad}$.

- **Net National Product (NNP):**
 - Adjusted GNP for depreciation.
 - $NNP = GNP - \text{Depreciation allowance}$.

- **National Income (NI):**
 - Total income earned by citizens resulting from ownership of resources used in production.
 - NI is often referred to as NNP.

- **Real GDP vs. Nominal GDP:**
 - Real GDP: Total market value of goods and services produced measured in constant prices.
 - Nominal GDP: Total market value measured in current prices.
 - Real GDP adjusts for inflation, while nominal GDP includes actual prices.

- **Price Deflator:**
 - GDP deflator: Ratio of nominal GDP to real GDP.
 - Measures economy's average price level.
 - Used to move from nominal to real GDP.

- Per Capita GDP:

- Total GDP divided by the population.
- Compares average income levels across countries.

- Purchasing Power Parity (PPP):

- Recognizes that income in one country may not purchase the same quantity of goods in another.
- PPP GDP per capita is used for comparison across countries at different levels of development.

- Per Capita Income, Personal Income, and Disposable Income:

- Per capita income: Average annual income per person.
- Personal income: Total income received by an individual from various factors of production.
- Disposable income: Personal income minus direct taxes.

- Drawbacks of GDP-Based Measures:

- Excludes informal and illegal economic activities.
- Ignores externalities like environmental pollution or resource depletion.
- GDP growth may not reflect true living standards or welfare if it comes at the cost of environmental degradation or other long-term risks.

Understanding and addressing the limitations of GDP-based measures is essential for accurately assessing the economic well-being and sustainability of a nation.

Lesson 30 Macroeconomic Equilibrium: The Determination of Equilibrium Income

Macroeconomic Equilibrium and Circular Flow:

Key Variables:

- Y: Income
- C: Consumption
- S: Savings
- I: Investment
- T: Taxes
- G: Government Expenditures
- M: Imports
- X: Exports

- Circular Flow:

- Illustrates the continuous movement of production, income, and resources between producers and consumers.
- Income flows from firms to households for factor services, then back to firms for goods and services.

- Leakages and Injections:

- Leakage: Use of household income not returned to domestic firms (e.g., savings, taxes, imports).
- Injection: Expenditure on GDP not originating from households (e.g., government spending, investment, exports).
- Injection-Leakage model used in Keynesian economics to identify equilibrium aggregate output.

- Macroeconomic Equilibrium:

- **Classical View:** S must equal I, G must equal T, and X must equal M.
- **Keynesian View:** Equilibrium when total injections equal total leakages or aggregate supply equals aggregate demand.

- Aggregate Demand and Supply:

- Aggregate demand (AD): Total planned spending in the economy.
- Aggregate supply (AS): Total value of goods and services firms would produce.
- Keynes focused on AD and AS equilibrium rather than individual market equilibriums.

- Consumption Function:

- Consumption (C): Amount of national income spent on domestic goods and services.
- Consumption function: $C = a + b(Y_d)$, where a is the minimum consumption level and b is the marginal propensity to consume (MPC).
- MPC: Extra amount consumed when disposable income increases by one dollar.
- MPS: Fraction of additional disposable income saved ($MPS = 1 - MPC$).

- Saving Function:

- Saving (S): Amount of saving households undertake at each income level.
- Saving function: $S = c + d(Y_d)$, where d is MPS.
- Relationship between saving and interest rate is positive.

- Calculation of APC, APS, MPC, MPS:

- APC: Total consumption divided by total disposable income.
- APS: Total saving divided by total disposable income ($APS = 1 - APC$).
- MPC: Change in consumption divided by change in disposable income.
- MPS: Change in saving divided by change in disposable income ($MPS = 1 - MPC$).

Lesson 31 Macroeconomic Equilibrium: The Determination of Equilibrium Income (Continued)

Investment and Investment Demand Curve:

- **Investment (I):** Economic activity that sacrifices current consumption for future output. It's the most volatile component of aggregate demand.
- **Investment Demand Curve:** Shows the negative relationship between investment level and the cost of borrowing (interest rate). Higher interest rates decrease investment demand.

Types of Investment:

1. **Residential Construction:** Depends on factors like house-buying willingness, wealth, ability to obtain loans, and housing costs.
2. **Non-residential Construction:** Influenced by firms' willingness to buy commercial property, vacancy rates, business needs, and rental costs.
3. **Producer Durables:** Depends on existing capacity utilization, technology availability, sales, and competition.
4. **Business Inventories:** Depend on current and expected sales, inventory prices, and certainty of deliveries.

Imports, Exports, and Trade Balance:

- **Imports:** Goods and services produced in another country and consumed domestically.
- **Exports:** Goods and services produced domestically and consumed in another country.
- Factors influencing imports and exports include income levels, exchange rates, prices, tariffs, and quotas.
- **Trade Balance:** Excess of exports over imports. A negative trade balance is a trade deficit.

Determination of National Income:

1. The 450 Line Approach to Equilibrium:

- Equilibrium occurs when aggregate demand (AD) equals aggregate income (Y).
- Shifts in AD affect income. For example, a \$10mn increase in government spending could lead to a \$50mn increase in income.
- Inflationary or deflationary gaps can occur if AD shifts from full employment equilibrium.

2. Algebraic Determination of Equilibrium:

- Equilibrium income formula: $Y = [1/(1-b)] \cdot [“a” + I + G + X - M]$, where b is the marginal propensity to consume (MPC).
- Equilibrium analysis can also be done using the injections-leakages approach, where withdrawals (S + M + T) equal injections (I + G + X).

3. Keynesian Multiplier:

- The Keynesian multiplier ($[1/(1-b)]$) represents the factor by which equilibrium output increases in response to changes in AD.
- The multiplier is larger with a higher MPC (b) and smaller with a lower MPC.

Understanding these concepts helps analyze the factors influencing investment, trade, and equilibrium income in an economy.

Lesson 32 Macroeconomic Equilibrium: The Determination of Equilibrium Income (Continued)

Keynesian AD & AS Approach to Equilibrium:

- **Aggregate Expenditures and Output:** As aggregate expenditures increase, both aggregate demand (AD) and output increase. The change in output is amplified due to the multiplier effect, especially when prices are constant (horizontal portion of AS).

Keynes's Intuition about the Multiplier:

- An increase in AD leads to higher wages, boosting consumption and production.
- Higher demand for labor raises incomes, further stimulating consumption and production.
- The multiplier effect is limited by leakages (savings, taxes, imports) from the circular flow.

Keynes Paradox of Thrift:

- Higher saving in a recession can worsen it. Increased saving reduces consumption and production, leading to lower incomes and potentially a recessionary spiral.

The Accelerator:

- The accelerator formalizes the investment response to output or income changes.
- Investment can rise rapidly during recovery, with the percentage increase in investment exceeding the percentage increase in income.
- Firms anticipate future demand growth and invest in capital accordingly, leading to substantial investment increases.

Interaction of Accelerator and Multiplier:

- Government spending increases can trigger a chain reaction of rising incomes and investment.
- However, this interaction is limited by the full-employment constraint and the need for sustained output growth.
- Investment is linked to changes in income, not the level of income, so sustained increases in income are necessary for continued investment growth.

Understanding these concepts helps analyze how changes in aggregate demand, saving behavior, and investment dynamics influence economic equilibrium and growth.

Lesson 33 The Four Big Macroeconomic Issues And Their Inter-Relationships ...

The Four Big Macroeconomic Issues and Their Inter-relationships:

1. Unemployment:

- **Importance:** Reduced national income, lost tax revenues, lower consumption, and increased social unrest.
- **Causes:** Historical context from the Great Depression to oil price shocks, technological shifts, and policy changes.
- **Policy Prescriptions:** Range from demand-side solutions (government spending) to supply-side (retraining, job matching).

2. Inflation:

- **Importance:** Erodes purchasing power, disrupts economic planning, and can lead to hyperinflation.
- **Causes:** External shocks (e.g., oil prices), excessive demand, cost-push factors, and monetary policy.
- **Policy Prescriptions:** Tightening monetary policy, reducing government spending, wage controls, and supply-side measures.

3. Balance of Payments Problem:

- **Importance:** Currency devaluation, debt crises, and reduced international competitiveness.
- **Causes:** Trade imbalances, capital flight, unsustainable borrowing, and exchange rate misalignment.
- **Policy Prescriptions:** Devaluation, fiscal austerity, import/export controls, and structural reforms.

4. Lack of Growth:

- **Importance:** Limits improvements in living standards, increases inequality, and undermines social cohesion.
- **Causes:** Low productivity, inadequate investment, political instability, and external shocks.
- **Policy Prescriptions:** Infrastructure investment, education and training, innovation incentives, and stable macroeconomic policies.

Inter-relationships:

- Macroeconomic problems are often interconnected. For example, high unemployment can reduce consumer demand, leading to deflationary pressures, while inflation can erode purchasing power, affecting consumer confidence and spending.
- Policies aimed at addressing one issue may inadvertently impact others. For instance, expansionary fiscal policy to combat unemployment may worsen inflation or exacerbate balance of payments deficits.
- Understanding these inter-relationships is crucial for policymakers to develop comprehensive strategies that mitigate the adverse effects of macroeconomic challenges while promoting sustainable economic growth.

Lesson 34 The Four Big Macroeconomic Issues And Their Inter-Relationships (Continued)

Hysteresis:

- Hysteresis refers to the permanent effects of a temporary change.
- In the context of unemployment, temporary demand falls can lead to longer spells of unemployment due to worker behavior and firm recruitment patterns.

Inflation and Deflation:

- Inflation is a continuous rise in the general price level, while deflation is the opposite.
- The rate of inflation is the percentage annual increase in the average price level.
- Ideal inflation rates vary but are generally positive, with rates above 10% considered undesirable.

- Various price indices like CPI and PPI measure inflation.

Costs of Inflation:

- Inflation redistributes income, increases uncertainty, leads to balance of payments problems, and causes resource wastage.

Theories about the Causes of Inflation:

- 1. Traditional Keynesian View:** Inflation and unemployment are seen as opposite sides of the same coin, caused by excessive aggregate demand (demand-pull inflation).
- 2. Cost-Push Inflation:** Arises from supply-side shocks that increase production costs, leading to higher inflation and unemployment (stagflation).

Lesson 35 The Four Big Macroeconomic Issues And Their Inter-Relationships (Continued)

Monetarists View:

- Monetarists attribute inflation to changes in the quantity of money (Quantity Theory of Money).
- They advocate for stable money supply growth aligned with the natural rate of output to control inflation.
- Monetarists view the Phillips curve as vertical in the long term, implying no trade-off between inflation and unemployment.

Balance of Payments (BOP):

- BOP records a country's transactions with the rest of the world.
- It includes exports, imports, financial capital flows, and transfers.

- BOP is a major indicator of a country's international trade status.

Market for Foreign Exchange:

- Functions like any other commodity market, with supply and demand determining exchange rates.
- Exchange rate specifies how much one currency is worth in terms of another.
- Pakistan's foreign exchange regime has evolved from linking the rupee to Pound Sterling to maintaining a floating rate system.

Lesson 36 The Four Big Macroeconomic Issues And Their Inter-Relationships (Continued.)...

Devaluation:

- Reducing a nation's currency exchange rate to raise exports' prices and lower imports' prices, thus boosting domestic production.
- Achieved by actively selling the currency and buying foreign currencies in the foreign exchange market.

Revaluation:

- Increasing a nation's currency exchange rate to lower exports' prices and raise imports' prices, appropriate when there's an undesired trade surplus.
- Done by buying the nation's currency and/or selling foreign currencies in the foreign exchange market.

Appreciation:

- Permanent increase in value or price, indicating greater consumer satisfaction and value.

- Common examples include real estate, art, stocks, and currency, often linked to a growing and healthy economy.

Depreciation:

- Permanent decrease in value or price due to reduced consumer satisfaction.
- Examples include capital, real estate, stocks, and currency, reflecting the economy's ability to produce goods.

Forces of Demand and Supply in Foreign Exchange Market:

- Supply curve shifts with transactions bringing dollars into the country (e.g., US investment, Pakistani exports).
- Demand curve shifts with transactions causing demand for dollars (e.g., Pakistani imports from the US, travelers to the US).

Equilibrium in the Market of Foreign Exchange:

- Occurs at the intersection of supply and demand curves, balancing positive and negative transactions.
- Under fixed exchange rates, government intervention is required to maintain equilibrium; under floating rates, market forces adjust the exchange rate.

Parts of BOP:

- 1. Current Account:** Records trade balance (exports - imports) and net factor receipts from abroad.
- 2. Capital Account:** Provides a picture of a country's net asset position, indicating increasing or decreasing indebtedness to the rest of the world.

3. Changes to Reserves: Reflects government interventions in the foreign exchange market, either building up or drawing down reserves to maintain balance.

Lesson 37 THE FOUR BIG MACROECONOMIC ISSUES AND THEIR INTER-RELATIONSHIPS (CONTINUED)

Determinants of the Capital Account:

- **Macroeconomic Environment:** A favorable macroeconomic environment, including stability and law and order, attracts foreign investment.
- **International Borrowing Conditions:** Foreign lenders' perception and interest rates influence a country's ability to raise foreign debt.
- **Interest Rate Differentials:** If foreign interest rates are lower than domestic rates, foreign investors may invest in the domestic market, affecting capital flows.

Competitiveness of Goods: Real Exchange Rate (RER):

- **Definition:** RER is the price of a domestic good relative to a foreign good, adjusted for nominal exchange rates and price levels.
- **Effect on Competitiveness:** When RER falls, domestic goods become relatively cheaper, improving competitiveness and potentially reducing current account deficits.
- **Interest Parity Condition:** An arbitrage condition stating that returns from investing in different currencies should be equal after accounting for interest rates and expected depreciation.
- **Current Account Deficit:** It's crucial for the long-term sustainability of the balance of payments; excessive deficits can strain an economy and lead to debt problems.

How to Reduce Current Account Deficit:

- **Devaluation:** Reducing the value of domestic currency can improve exports and reduce imports, potentially lowering the current account deficit.

- **Challenges:** Devaluation may not always be effective due to various factors, and other policies may be needed to address deficits effectively.

Lesson 38 The Four Big Macroeconomic Issues And Their Inter-Relationships (Continued)...

The Concept of Economic Growth and Growth Rate:

- Economic growth refers to an increase in an economy's level of production, output, or income.
- Growth rate of real GDP can be negative, positive, or zero, with normal rates ranging from 2-3% for mature developed countries and 5-7% for less developed countries.

Actual & Potential GDP:

- The GDP gap or output gap is the difference between actual GDP and potential GDP.
- Positive gap indicates expansion, negative gap indicates recession.
- Percentage GDP gap = $(\text{Actual GDP} - \text{Potential GDP}) / \text{Potential GDP}$.

Traditional Thinking About Growth:

- Growth can result from increased factor resources or more efficient factor utilization.
- Policy implications: accumulate factors of production or develop cost-efficient technologies.

Real vs Nominal GDP:

- Nominal GDP uses current prices; real GDP adjusts for inflation.
- Real GDP is used for year-over-year GDP growth comparisons.

Aggregate GDP vs Per Capita Real GDP:

- Aggregate GDP is total income of an economy in a period; per capita real GDP adjusts GDP for population.
- Per capita real GDP growth rate gives a more accurate indication of living standard improvements.

Link Between Growth and Factors of Production:

- **Capital:** Increasing capital can boost growth, measured by capital deepening or capital widening.
- **Labor:** Quantity and quality of labor affect growth.
- **Land:** Efficient use of land can drive growth.
- **Raw materials:** Adequate raw materials increase output.
- **Technical knowledge:** Technological advancements drive production and growth.

Lesson 39 The Four Big Macroeconomic Issues And Their Inter-Relationships (Continued)...

Exogenous Growth Theory:

- Also known as the Neo-classical or Solow growth model.
- Developed by Robert Solow, it assumes diminishing returns to capital.
- The model suggests that at some point, new capital only replaces existing capital lost due to depreciation, halting economic growth.

Mathematical Form of the Model:

- Cobb Douglas production function is often used.

- Per capita production function: $y = k^\alpha$, where y is per capita output, k is per capita capital, and α is a parameter between 0 and 1.
- Capital accumulation and steady state level of capital are key concepts in the model.

Inferences from the Theory:

- Saving has no effect on the steady state growth rate.
- Determinants of growth are beyond policymakers' control.
- Convergence theory suggests countries with higher capital levels converge to the output level of countries with lower capital.

Criticism / Weaknesses:

- Empirical evidence offers mixed support.
- Limitations include neglect of entrepreneurship, institutional strength, and explanation of technological progress.
- Endogenous growth theory addresses these weaknesses.

Endogenous Growth Theory:

- Developed in the 1980s as a response to criticism of the exogenous growth model.
- Macro models built from micro foundations.
- Emphasizes the role of new technologies and human capital in driving growth.
- Policy measures can influence long-run growth rate, unlike in exogenous growth theory.

Labor, Capital, and Land:

- Labor quality and quantity, capital accumulation, and technological progress are engines of growth.
- Human capital quality can increase output even with limited physical capital.
- Technological progress is cumulative and depends on invention, innovation, and learning.
- Natural resources like land and raw materials play a role, but technological progress is more significant for sustained growth.

Lesson 40 The Four Big Macroeconomic Issues And Their Inter-Relationships: Bilateral Relationship among the “Big Four” and Fiscal Policy

Short-Run Big Four Objectives:

- In the short term (up to about two years), the big four objectives are faster growth, lower unemployment, lower inflation, and avoidance of balance of payments deficits.
- These objectives are related and depend on aggregate demand, varying with the trade cycle.

Long-Run Big Four Inter-relationships:

- In the long run, the relationships between inflation, unemployment, growth, and balance of payments can be complex.
- Various relationships such as inflation-unemployment, growth-inflation, growth-unemployment, and growth-balance of payments are discussed.

The Salter-Swan Diagram:

- This framework allows the analysis of relationships between macroeconomic problems and policy prescriptions.

- It uses exchange rate-absorption space to analyze internal and external balance.

Fiscal Policy:

- Fiscal policy is the use of government spending and taxation to stabilize the business cycle.
- It includes government purchases, taxes, and public debt management.
- The consequences of fiscal policy are observed in terms of the federal deficit.

Types of Taxes:

- Direct taxes are levied directly on income, while indirect taxes are imposed on goods and services.
- Taxes can be regressive, proportional, or progressive, depending on how they affect different income levels.

The Laffer Curve:

- It shows the relationship between tax rates and total tax revenue.
- There's an optimum tax rate that maximizes revenue, lying between 0% and 100%.

Effect of Taxation on Equilibrium Output and the Balanced Budget Multiplier:

- Taxes act as a drag on the multiplier effect of government spending.
- The tax-adjusted multiplier is smaller than the basic Keynesian multiplier.
- The balanced budget multiplier can still have a positive effect, though smaller than without taxation.

Overall, the lesson covers various aspects of macroeconomic issues and fiscal policy, including their interrelationships and effects.

Lesson 41 Fiscal Policy, Money and Banking

Assets:

- 1. Loans to Government:** Provided to finance government expenditures during budget deficits.
- 2. Notes, Coins & Currency in Circulation:** Responsible for issuing and regulating physical currency.
- 3. Forex Reserves:** Used to stabilize domestic currency exchange rates and intervene in foreign exchange markets.
- 4. Government & Commercial Bank Deposits:** Deposits held by governments and commercial banks for various purposes, including meeting reserve requirements and facilitating transactions.
- 5. Loans to Private Sector:** Extended to private sector entities for liquidity support or addressing specific sectoral needs.

Liabilities:

- 1. Liquidity Paper Issued:** Short-term liquidity instruments like treasury bills or central bank bills issued to manage banking system liquidity and implement monetary policy.

Additional points:

- The balance sheet composition may vary based on the central bank's functions, monetary policy objectives, and economic context.

- Central banks may hold other assets and liabilities like gold reserves or special drawing rights (SDRs).
- The balance sheet provides insights into a central bank's role in the economy, monetary policy operations, and financial stability.

Lesson 42 Money, Central Banking and Monetary Policy

Balance Sheet of State Bank of Pakistan (SBP)

Assets:

- 1. Forex Reserves:** Foreign exchange reserves including foreign currencies, gold, and silver reserves.
- 2. Loans/Credit to Government:** Financing provided to the government, including outstanding treasury bonds and bills.
- 3. Loans/Credit to Private Sector:** Advances extended to commercial banks or private sector entities.

Liabilities:

- 4. Notes, Coins & Currency in Circulation (M0):** Physical currency in circulation, considered a liability for the central bank.
- 5. Government & Commercial Bank Deposits:** Positive account balances held by commercial banks and/or the government.

6. Liquidity Paper Issued: Bills issued by the central bank to manage liquidity in the financial system.

Monetary Policy Instruments:

- 1. Reserve Ratio and SLRs:** Central bank sets and alters mandatory reserve ratios for commercial banks, affecting the money multiplier and statutory liquidity requirements.
- 2. Discount Rate:** Rate at which central bank lends to commercial banks, influencing banks' reserve ratios and money multiplier.
- 3. Open Market Operations (OMOs):** Buying or selling government securities to affect money supply directly.
- 4. Foreign Exchange Market Interventions:** Purchasing or selling foreign exchange, impacting foreign exchange reserves and consequently money supply.

Functions of Central Bank:

- Lender of last resort, bailing out commercial banks facing liquidity shortfalls.
- Supervisor of the financial system, ensuring stability and monitoring risk-taking, capital adequacy, and liquidity positions of financial institutions.
- Intervener in the foreign exchange market, responsible for exchange rate policy and balance of payments.

Motives for Holding Money:

- **Transactions Motive:** Holding cash for day-to-day transactions, influenced by income levels.
- **Precautionary Motive:** Holding cash for unexpected expenses or investment opportunities.
- **Assets Motive:** Holding cash to switch between various investments, influenced by income and interest rates.

Demand for Money:

- Money demand increases with real income levels and falls with real interest rates.
- Real money supply depends on the assumption regarding price levels.

This comprehensive understanding of the balance sheet components, monetary policy instruments, central bank functions, and motives for holding money provides insights into the functioning of the State Bank of Pakistan and the broader monetary system.

Lesson 43 Money and Goods Market Equilibrium: IS-LM Framework Money Supply and

Equilibrium in the Money Market:

- **Money Demand (L):** Increases with income and decreases with interest rates, resulting in a downward-sloping line in i - M (interest rate vs. real money) space.
- **Real Money Supply (M_s/P):** Controlled by the central bank, represented as a vertical line intersecting the M axis at a finite point determined by M_s and the price level (P).
- **Equilibrium:** Intersection of the money demand (L) and real money supply (M_s/P) curves, determining the equilibrium interest rate.
- **Shifts in Equilibrium:** Changes in income (Y) cause shifts in the money demand curve, impacting equilibrium interest rates.

The LM Curve:

- **Definition:** Captures combinations of real output and real interest rate at which the money market is in equilibrium.
- **Shifts:** Any change in real money supply (M_s/P) causes shifts in the LM curve, with an increase causing the curve to shift down (or to the right), and a decrease causing it to shift up (or to the left).

Equilibrium in the Goods Market: The IS Curve:

- **Investment Schedule (I):** Plots as a downward-sloping curve in i - S (interest rate vs. saving) space.
- **Effect of Interest Rate Changes on Investment:** Lower interest rates lead to higher investment demand, resulting in a rise in equilibrium income.
- **Slope of the IS Curve:** Depends on the sensitivity of investment demand to interest rate changes (ieI) and the Keynesian multiplier (k).

Shifts in the IS Curve:

- Any autonomous changes in consumption (C), government spending (G), or net exports ($X - M$) cause shifts in the IS curve.

Joint Equilibrium in Goods and Money Markets: The IS-LM Framework:

- Intersection of the upward-sloping LM curve and the downward-sloping IS curve represents joint equilibrium.

- Analysis of fiscal and monetary policy interactions to guide the economy towards full-employment equilibrium.
- Evaluation of the effectiveness of fiscal vs. monetary policy based on the slopes of the IS and LM curves: Fiscal policy is more effective when the LM curve is relatively flat, while monetary policy is more effective when the IS curve is relatively flat.

Understanding the IS-LM framework allows economists to study how changes in fiscal and monetary policy affect equilibrium in both the goods and money markets, and how these policies can be coordinated to achieve desired macroeconomic outcomes.

Lesson 44 International Trade And Finance

International Trade:

- **Importance:** Influences balance of payments (BOP) and foreign exchange reserves, determines production and consumption possibilities, affects aggregate demand, income, and employment.
- **Interesting Facts:**
 1. World trade value increased 20-fold from 1930-2000.
 2. Contribution of exports to GDP doubled on average from 30% to 50%.
 3. Shift from 50%-50% to 75%-25% in favor of manufactured goods in world exports.
 4. 50% of world trade between high-income countries (HICs), 14% between low-income countries (LICs), rest between both.
- **Reasons for Trade:** Mutual gains from trade, realized through comparative advantage.

Comparative Advantage:

- **Definition:** Countries gain by specializing in and exporting goods they have a comparative advantage in.
- **Illustration:** Example of US and UK production of cotton and food demonstrates comparative advantage.
- **Source:** Productivity differentials (Ricardo) or differences in factor endowments (Heckscher-Ohlin).

Criticism against Heckscher-Ohlin Theory:

Static View: Views comparative advantage statically, critics argue for dynamic view.

- **Dynamic Comparative Advantage:** Countries can develop comparative advantage in different goods over time.

Welfare Effects of Tariffs:

- **Revenue Generation:** Tariffs generate revenue but may harm domestic consumers by protecting less efficient domestic producers.

International Finance:

- Concerned with mobility of financial capital across countries, includes issues like exchange rate regimes and balance of payments (BOP) crises.
- **Types of Transactions:** Foreign direct investment (FDI), foreign portfolio investment (FPI), debt flows, and aid flows.
- **Growth in Private Capital Flows:** Phenomenal growth since the 1990s, reflecting financial market integration and capital account liberalization.

Benefits of International Capital Mobility:

- Consumption smoothing, risk diversification, fiscal policy effectiveness.

Disadvantages of International Capital Mobility:

- Ineffective monetary policy, pro-cyclical capital flows exacerbating economic fluctuations, volatility leading to speculation and crises.

Suggestions to Cure Problems of Global Capital Mobility:

- Strengthening recipient countries' financial and macroeconomic fundamentals, reforming international financial architecture, implementing tax-like controls on capital movements.

Understanding international trade and finance is crucial for policymakers and economists to address issues related to economic growth, stability, and development in an increasingly interconnected global economy.

Lesson 45 Problems of Lower Income Countries (LICs)

International Trade:

- **Importance:** Influences balance of payments (BOP) and foreign exchange reserves, determines production and consumption possibilities, affects aggregate demand, income, and employment.

- **Interesting Facts:**

1. World trade value increased 20-fold from 1930-2000.
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Let's ace those studies together!

Feel free to ask me about any queries

Remember me in your prayers

☆☆☆☆☆ Wish you all the best ☆☆☆



Mr Shāhīzāz